

Beauty in Context: A Unifying Mechanistic Framework for the Cheerleader, Star, and Costume Effects and Their Cross-Sex Asymmetries

(expanded biological, sociological, and neuroendocrine treatment)

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Abstract

Perceived physical attractiveness is often treated as a stable property of a face or body, yet it is systematically modulated by social context. We integrate three such modulators under a single mechanistic framework: the *cheerleader effect* (greater attractiveness in a group than alone), the *star effect* (greater attractiveness when at the focal centre of social attention—distinct from the egocentric *spotlight effect*), and the *costume effect* (greater attractiveness when dress is incongruent with the local norm). We argue that all three are special cases of three mechanisms—ensemble/statistical perception, social-proof/status inference, and salience/distinctiveness $\times$ arousal—and, new to this version, we trace each mechanism down three levels of explanation: an *evolutionary/biological* level (why the cue is informative about mate value), a *neuroendocrine* (“chemical”) level (the reward, arousal, and hormonal systems that implement it), and a *sociological* level (status, norms, and social proof). We analyse each effect bidirectionally (male perceivers judging female targets, $M \rightarrow F$; female perceivers judging male targets, $F \rightarrow M$) and predict approximate symmetry for the cheerleader effect, a larger $F \rightarrow M$ asymmetry for the star effect, and a smaller, non-monotonic $M \rightarrow F$ pattern for the costume effect. To make the predictions concrete we translate them onto a 0–10 rating scale: a target of average attractiveness (5.0) is predicted to be seen as roughly 5.4–5.7/10 under these effects, the largest single bump ($\approx +0.7$ points) being the star effect for women judging men. Two clearly labelled simulations illustrate the mechanisms. The cheerleader effect is established; the star and costume effects are presented as novel, falsifiable framings.

Keywords: attractiveness; mate choice; ensemble perception; mate-choice copying; social proof; reward; testosterone; arousal; sex differences.

1 Introduction

A long tradition in person perception treats physical attractiveness as a property carried by the stimulus: a face has features—averageness, symmetry, sexual dimorphism—that elicit a relatively fixed aesthetic response (Langlois et al., 2000; Langlois & Roggman, 1990; Rhodes, 2006; Thornhill & Gangestad, 1999). This view captures real regularities, but it is incomplete. The same face can be rated differently depending on *who else is in the frame*, *where attention is pointing*, and *what the person is wearing relative to the crowd*. The central puzzle of this paper is that perceived attractiveness is not a scalar attached to a face but the output of an inference that takes social context as input.

We focus on three context-dependent modulators. First, the **cheerleader effect**: individuals are rated as more attractive when seen as part of a group than when seen alone (Walker & Vul, 2014). Second, what we will call the **star effect**: a person at the focal centre of social attention—the celebrant at a

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birthday party, the graduate on stage, the protagonist of a story—is perceived as more attractive than the same person off-centre. Third, the **costume effect**: a person whose dress breaks the local norm (cosplay at a commuter station, formalwear at a beach) is perceived as more attractive than the same person in norm-congruent dress. The first effect is empirically established; the second and third, as defined here, are largely novel framings that we advance as conjectures to be tested, and we flag them as such throughout.

1.1 Three levels of explanation

A recurring weakness of “effect” catalogues in social psychology is that they stop at the phenomenon. This version of the paper deliberately pushes each mechanism down three explanatory levels, because a context effect on attraction is only convincing if we can say why it should exist, how the brain and body implement it, and how it rides on social structure.

Biological (ultimate) level. Why would a perceptual or social cue be *informative* about mate value, such that selection would favour attending to it? Attractiveness cues such as averageness and symmetry plausibly signal developmental stability and heritable quality (Little, Jones, & DeBruine, 2011; Rhodes, 2006; Thornhill & Gangestad, 1999), and attending to *others’* choices (social proof) is an adaptive shortcut when quality is hard to assess directly (Gouda-Vossos, Nakagawa, Dixson, & Brooks, 2018; Place, Todd, Penke, & Asendorpf, 2010).

Neuroendocrine (“chemical”) level. How is the valuation physically implemented? Viewing attractive faces engages dopaminergic reward circuitry—orbitofrontal cortex, ventral striatum/nucleus accumbens (Aharon et al., 2001; O’Doherty et al., 2003; Winston, O’Doherty, Kilner, Perrett, & Dolan, 2007)—and the reward response shows sex differences (Cloutier, Heatherton, Whalen, & Kelley, 2008). Arousal systems (sympathetic activation, mis-attributable to attraction; Dutton & Aron, 1974; Schachter & Singer, 1962) and gonadal hormones (testosterone and status; oestrogen-linked cyclic shifts) supply additional, if contested, machinery (Gildersleeve, Haselton, & Fales, 2014; Mazur & Booth, 1998; Wood, Kressel, Joshi, & Louie, 2014).

Sociological level. How does the cue ride on social structure? Attractiveness functions as a diffuse status characteristic (Webster & Driskell, 1983); attention and deference track prestige (Henrich & Gil-White, 2001); desire for status is itself a candidate fundamental motive (Anderson, Hildreth, & Howland, 2015); and dress is a normative signalling system whose violation is socially legible.

We invoke these levels for each effect below. We are candid that the biological and (especially) neuroendocrine levels are largely *correlational substrates*: mapping them onto these specific context effects is, at present, theoretical scaffolding rather than demonstrated fact, and we hedge accordingly.

1.2 Bidirectionality

Because attraction research is shaped by robust sex differences in mate preferences and mating psychology (Buss, 1989; Buss & Schmitt, 1993; Trivers, 1972), every effect must be examined in both directions of heterosexual judgement—male perceivers judging female targets ($M \rightarrow F$) and female perceivers judging male targets ($F \rightarrow M$)—asking whether the effect is symmetric or asymmetric across perceiver sex, and *why*. We use “sex” throughout as the design factor defining perceiver and target categories; this framing is a simplification that brackets same-sex attraction, bisexual perceivers, and non-binary identities, to which we return in the Limitations. We close with a deliberately concrete payoff (Section 7): the predicted change, in points on a 0–10 scale, for a target who starts at the average.

2 The cheerleader effect

2.1 Definition and evidence

Walker and Vul (2014) reported that the same face is rated as more attractive when presented within a group of other faces than when presented in isolation—a small but replicable boost they attributed to *hierarchical encoding*. On their account, the visual system summarises a set of faces by an ensemble average and biases the representation of each member toward that average; because average faces are themselves attractive (Langlois et al., 2000; Langlois & Roggman, 1990), regression of each face toward the group mean tends to raise its perceived attractiveness. A related but distinct phenomenon, the *group attractiveness effect*, shows that the rated attractiveness of a *group as a whole* can exceed the mean attractiveness of its members (van Osch, Blanken, Meijs, & van Wolferen, 2015).

The cheerleader effect is real but bounded, and its mechanism is contested. Ying, Burns, Lin, and Xu (2019) provided evidence consistent with an ensemble/face-adaptation account. In contrast, Carragher, Thomas, Gwinn, and Nicholls (2019) found only limited evidence that the boost depends on hierarchical (perceptual) encoding, and Hsieh et al. (2021) argued that the effect reflects a *memory bias* rather than online perceptual averaging. We therefore treat the *existence* of a modest group-boost as established while treating its *mechanism* as open.

2.2 Mechanism across three levels

Biological level. Ensemble (summary-statistical) perception is a general, capacity-efficient feature of vision: observers rapidly extract the average of a set when they cannot encode every member (Haberman & Whitney, 2007; Whitney & Yamanashi Leib, 2018). The reason averaging *raises* attractiveness is that averageness is itself attractive, plausibly because an average face is a low-mutation-load, developmentally stable template (Langlois & Roggman, 1990; Little et al., 2011; Thornhill & Gangestad, 1999). The cheerleader effect is thus a by-product of an efficient perceptual heuristic colliding with an evolved preference for the average.

Neuroendocrine (“chemical”) level. Attractive faces are processed as *rewards*: they recruit orbitofrontal cortex and ventral striatal reward circuitry, and people will expend effort to view them (Aharon et al., 2001; O’Doherty et al., 2003; Winston et al., 2007). If ensemble encoding nudges a face toward the attractive average, the prediction is a small upward shift in this reward signal. Crucially for our bidirectional analysis, the reward response to facial attractiveness is not identical across perceiver sex (Cloutier et al., 2008); however, because ensemble averaging is an *early perceptual* operation upstream of valuation, we expect the sex-dependence it inherits to be modest (see below).

Sociological level. The cheerleader effect is the most “perceptual” and least “social” of the three; its sociological content is limited to the fact that humans are usually encountered in groups, so the ensemble prior is ecologically the default rather than the exception.

2.3 $M \rightarrow F$ versus $F \rightarrow M$

Ensemble coding is an early, domain-general visual operation; there is no strong a priori reason that the averaging operation itself should differ by perceiver sex, so we predict *approximate symmetry*. One downstream asymmetry is worth naming: a *ceiling* consideration. If the female-target attractiveness distribution that male perceivers work with is rated nearer the top of the scale, there is less headroom for an upward group-boost, which would make the cheerleader effect *slightly larger for male targets* ($F \rightarrow M$)—an asymmetry driven by measurement ceiling rather than by the mechanism. We flag this as a measurement-level caveat, not a mechanistic asymmetry. Either way, the near-symmetry of the perceptual

route is a useful contrast case: whatever asymmetries we find for the star and costume effects are then unlikely to be artefacts of a generic group-viewing confound.

3 The star effect

3.1 Definition and a necessary disambiguation

We define the **star effect** as the increase in perceived attractiveness of a person who occupies the focal centre of social attention relative to the same person in a non-focal position. Paradigm cases: the person others orient toward at a party, the performer on a lit stage, the named protagonist of a narrative.

This must be sharply distinguished from the established *spotlight effect* of Gilovich, Medvec, and Savitsky (2000), which concerns *self*-perception: the egocentric tendency to overestimate how much one's own appearance and behaviour are noticed by others. The spotlight effect is about the actor's inflated sense of being observed; the star effect, as we use it, is about an *observer's* upgraded evaluation of a target who is in fact attended to by others. We adopt "star effect" to avoid this conflation; the phenomenon is, to our knowledge, a novel framing rather than an established labelled effect.

3.2 Mechanism across three levels

Biological level. The functional logic is *mate-choice copying*: when assessing mate value is costly or uncertain, copying the revealed preferences of others is an adaptive information shortcut. People find a target more desirable after observing that others have chosen or attend to that target (Little, Burriss, Jones, DeBruine, & Caldwell, 2008; Place et al., 2010; Waynforth, 2007), an effect summarised meta-analytically by Gouda-Vossos et al. (2018). Parental-investment asymmetries (Buss, 1989; Buss & Schmitt, 1993; Trivers, 1972) predict that the sex with higher obligatory investment (typically female) should weight such social information more—the seed of the star asymmetry.

Neuroendocrine ("chemical") level. Two systems are implicated. First, *reward*: directed gaze and attention from others is itself processed as rewarding, and the reward value of a face interacts with whether it is looking at (attending to) the viewer (Kampe, Frith, Dolan, & Frith, 2001); social endorsement can thus plausibly up-regulate the striatal/orbitofrontal reward signal that attractive faces already evoke (Aharon et al., 2001; Cloutier et al., 2008). Second, *status and testosterone*: occupying the centre of attention is a dominance/prestige display, and the biosocial model links status contests to testosterone dynamics in men (Mazur & Booth, 1998); on the perceiver side, women's attractiveness judgements of men track cues to men's testosterone and prosociality (Roney, Hanson, Durante, & Maestripieri, 2006), consistent with attention-as-status being especially potent in the F → M direction. (Oxytocin's role in face reward is real but pertains mainly to *pair-bonded* valuation rather than initial rating; Scheele et al., 2013—we mention it only to bound the claim.)

Sociological level. Sociologically, the star effect is the inverse of "beauty as status": if attractiveness confers status (Webster & Driskell, 1983), then conferred status (attention, deference) should be read back as evidence of underlying value via prestige (Henrich & Gil-White, 2001), the halo (Dion, Berscheid, & Walster, 1972; Thorndike, 1920), and social proof (Cialdini & Goldstein, 2004). The strength of this inference scales with how fundamental status concerns are to the perceiver (Anderson et al., 2015).

3.3 M → F versus F → M

We predict the star effect to be **larger F → M than M → F**. The convergent rationale across levels: mate-choice copying is more consistently observed in women (Gouda-Vossos et al., 2018); the weighting of status/resource cues is on average higher for female perceivers (Buss, 1989; Buss & Schmitt, 1993);

and women’s judgements of men are sensitive to status- and hormone-linked cues (Roney et al., 2006). We note the obvious confounds (attention can co-vary with dominance displays, physical attractiveness, or performance quality) that the proposed experiments are designed to control, and we present the direction as a conjecture.

4 The costume effect

4.1 Definition and evidence

We define the **costume effect** as the increase in perceived attractiveness produced by dress that is incongruent with the local norm—cosplay, a costume, or simply clothing that breaks the prevailing dress code. The integrated phenomenon is a novel framing, but its ingredients are documented. The most direct evidence is the *colour-red effect*: red clothing or background increases men’s attraction to women (Elliot & Niesta, 2008) and women’s attraction to men, where it is read partly through status (Elliot et al., 2010). The red effect is also a cautionary anchor: large pre-registered replications find it weak or absent (Lehmann & Calin-Jageman, 2017), underlining that sartorial influences on attraction are real but small and moderator-dependent.

4.2 Mechanism across three levels

Biological level. The core is *distinctiveness*: a stimulus that stands out from its context is preferentially encoded—the von Restorff isolation effect (Hunt, 1995). Norm-breaking dress makes the wearer “pop” from the local distribution, increasing individuation and encoding strength, which is a precondition for any further valuation.

Neuroendocrine (“chemical”) level. This is where “chemistry” is most literally apt. Novel or norm-violating stimuli elevate sympathetic *arousal* (catecholamines—adrenaline/noradrenaline—and autonomic activation). By the two-factor theory of emotion (Schachter & Singer, 1962), arousal is labelled by context, and under romantic framing it can be *mis-attributed* to attraction—the classic high-arousal bridge result (Dutton & Aron, 1974). The costume effect thus has a plausible arousal-transfer route: incongruous dress raises arousal, which inflates attraction ratings when the target is a candidate mate. Gonadal hormones may modulate the F → M side: women’s preferences for masculine/status cues shift across the ovulatory cycle in some analyses (Gildersleeve et al., 2014) though not others (Wood et al., 2014), so any cycle-linked amplification of responses to bold male display is offered only as a hedged possibility.

Sociological level. Sociologically, dress is a normative code, and its violation is a *signal*—of confidence, nonconformity, or status. Bold or rule-breaking dress can read as a costly, hence credible, status display (Anderson et al., 2015), and clothing also changes the *wearer’s* own cognition and bearing (enclothed cognition; Adam & Galinsky, 2012), an indirect route by which a costume raises attractiveness through altered behaviour. Because incongruity simultaneously costs *processing fluency* (which usually lowers liking), the net effect is a trade-off, predicting a *non-monotonic*, inverted-U relationship with degree of incongruity. The *pratfall effect* (Aronson, Willerman, & Floyd, 1966) is an analogue of this “optimal deviation” logic.

4.3 M → F versus F → M

We tentatively predict the costume effect to be *modestly larger M → F than F → M*, because the arousal/novelty channel is, on average, more tightly coupled to visual sexual interest in male perceivers, and because clothing-based display is a more conventionalised component of female self-presentation that male perceivers are attuned to. This is our weakest directional prediction: the red effect appears in

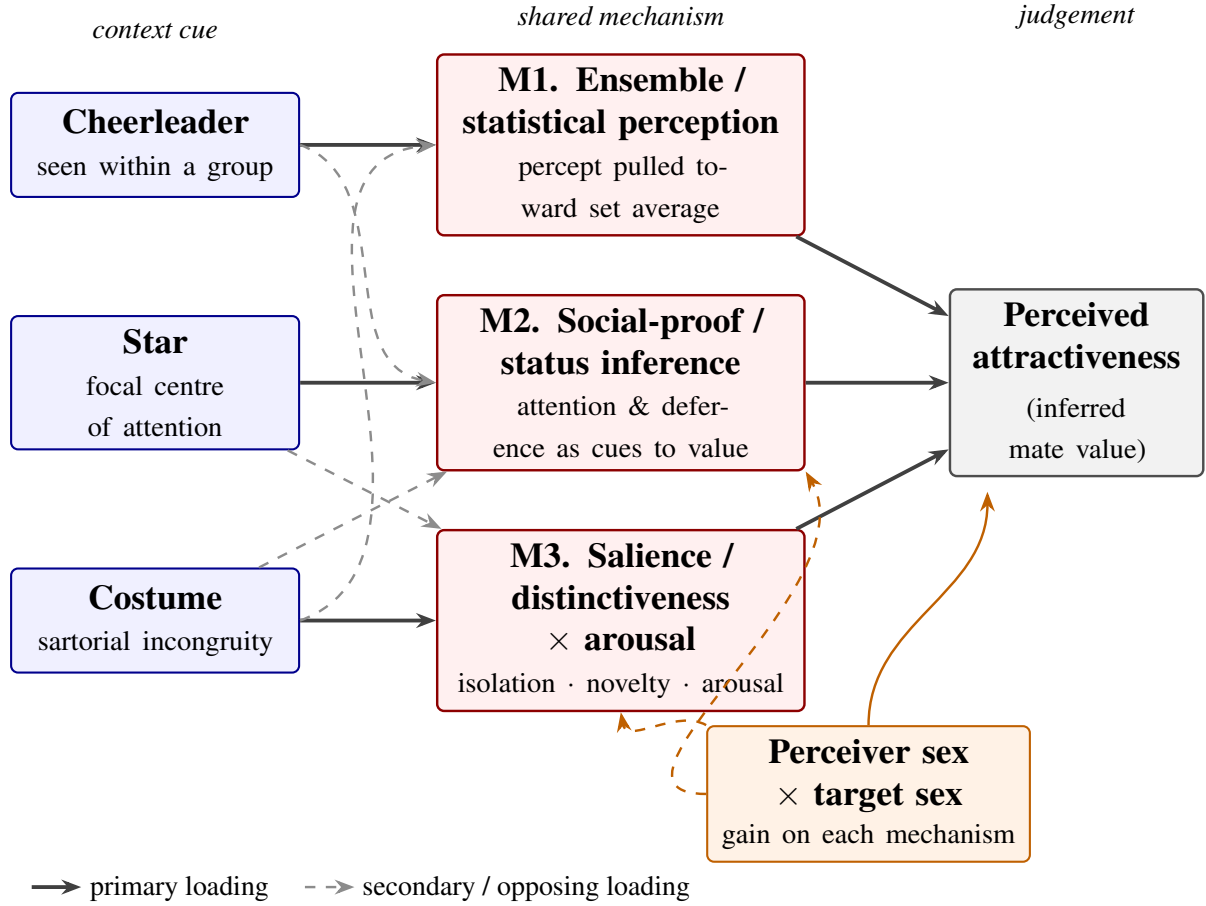


Figure 1. Conceptual diagram of the unifying framework (conceptual, not data). Three context cues (left) drive perceived attractiveness (right) through three shared mechanisms (centre). Solid arrows denote primary loadings; dashed arrows denote secondary or opposing loadings. Perceiver sex $\times$ target sex (orange) sets the gain on the mechanism $\rightarrow$ attractiveness mapping and is the locus of the predicted cross-sex asymmetries.

both directions (Elliot et al., 2010; Elliot & Niesta, 2008), and the status route to costume could equally amplify the $F \rightarrow M$ direction. We expect a smaller and more context-dependent asymmetry here than for the star effect.

5 A unifying framework

5.1 Three mechanisms, three effects

The three effects map onto three mechanisms (Figure 1): **M1, ensemble/statistical perception**; **M2, social-proof/status inference**; and **M3, salience/distinctiveness $\times$ arousal**. Each effect loads primarily on one mechanism (cheerleader $\rightarrow$ M1, star $\rightarrow$ M2, costume $\rightarrow$ M3) but secondarily on others, sometimes with opposing sign (costume loads M3 positively but M1 negatively, because incongruity disrupts ensemble fluency). The three explanatory levels of Section 1.1 cut *across* the mechanisms: each mechanism has an evolutionary rationale, a neuroendocrine substrate, and a sociological expression.

5.2 Semi-formal statement

Let a perceiver of sex s judge a target of sex t embedded in context c (one of the three effects, or a neutral baseline). Write the perceived attractiveness as a baseline plus mechanism-weighted contributions:

$$A(\text{target} \mid c, s, t) = a_0 + \sum_{m \in \{1,2,3\}} \underbrace{\lambda_{c,m}}_{\text{loading of effect } c \text{ on } m} \cdot \underbrace{g_{s,t,m}}_{\text{perceiver} \times \text{target gain}} \cdot \underbrace{\phi_m(c)}_{\text{mechanism activation}} + \varepsilon, \quad (1)$$

where a_0 is the context-free attractiveness of the face/body, $\lambda_{c,m} \in \mathbb{R}$ encodes how strongly (and with what sign) effect c engages mechanism m , $\phi_m(c) \geq 0$ is the activation of mechanism m in context c , $g_{s,t,m} \geq 0$ is the sex-dependent gain on mechanism m , and ε is noise. The contextual modulation—the “effect”—is the contrast $\Delta_c = A(\cdot \mid c) - A(\cdot \mid \text{baseline})$. For M3 we allow ϕ_3 to be non-monotonic (inverted-U) in the degree of incongruity. The neuroendocrine level gives $g_{s,t,m}$ a candidate physical meaning: it is the gain of the reward/arousal/status systems that read out mechanism m , and its sex-dependence is exactly what the bidirectional design estimates.

From Equation 1 we read off six predictions:

- P1 (existence/direction).** For each effect, $\Delta_c > 0$ over the typical operating range.
- P2 (cheerleader symmetry).** Because M1 is an early perceptual operation, $g_{s,t,1}$ is approximately constant in s ; hence $\Delta_{\text{cheer}}^{\text{M} \rightarrow \text{F}} \approx \Delta_{\text{cheer}}^{\text{F} \rightarrow \text{M}}$.
- P3 (star asymmetry).** Because social-proof/status cues are weighted more by female perceivers, $g_{\text{F} \rightarrow \text{M},2} > g_{\text{M} \rightarrow \text{F},2}$; hence $\Delta_{\text{star}}^{\text{F} \rightarrow \text{M}} > \Delta_{\text{star}}^{\text{M} \rightarrow \text{F}}$.
- P4 (costume non-monotonicity & mild asymmetry).** ϕ_3 is inverted-U in incongruity, and $g_{\text{M} \rightarrow \text{F},3} \gtrsim g_{\text{F} \rightarrow \text{M},3}$ via the arousal channel, so $\Delta_{\text{costume}}^{\text{M} \rightarrow \text{F}} \gtrsim \Delta_{\text{costume}}^{\text{F} \rightarrow \text{M}}$, with a smaller gap than P3.
- P5 (cross-effect ordering).** In attention-rich settings the star effect dominates and does so most for $\text{F} \rightarrow \text{M}$; in static visual-array settings the cheerleader and costume effects dominate and do so most for $\text{M} \rightarrow \text{F}$.
- P6 (sub-additive / antagonistic co-occurrence).** When two effects load on the same mechanism their combination is sub-additive; when they load with opposing signs (costume $\times$ cheerleader on M1) they partially cancel; and the cheerleader and star effects may be *antagonistic*, because directing a group’s attention onto one member individuates that member and defeats the ensemble-assimilation that drives the cheerleader boost.

Figure 2 traces each effect through its dominant mechanism to the attractiveness judgement and annotates the predicted cross-sex asymmetry.

6 Cross-effect and cross-sex comparison

Table 1 summarises the qualitative predictions, and Figure 3 renders illustrative predicted effect sizes. The magnitudes in Figure 3 are *a priori priors*—reasoned conjectures used to communicate the predicted pattern and to seed the power analysis—not estimates from data.

Three comparative claims follow. *Which effect is strongest?* No effect dominates unconditionally; the framework predicts a *context* $\times$ *perceiver-sex* interaction (P5). For $\text{F} \rightarrow \text{M}$, the star effect is predicted to be the largest single modulator (M2 up-weighted); for $\text{M} \rightarrow \text{F}$, the costume and cheerleader effects are comparatively more influential (perceptual/arousal channels weighted at least as heavily as social proof). *For whom is each effect strongest?* Star: women judging men. Costume: men judging women. Cheerleader: roughly equal. *Under what conditions?* The star effect requires a genuine attention cue and shrinks toward zero when attention is decoupled from value; the costume effect follows an inverted-U and

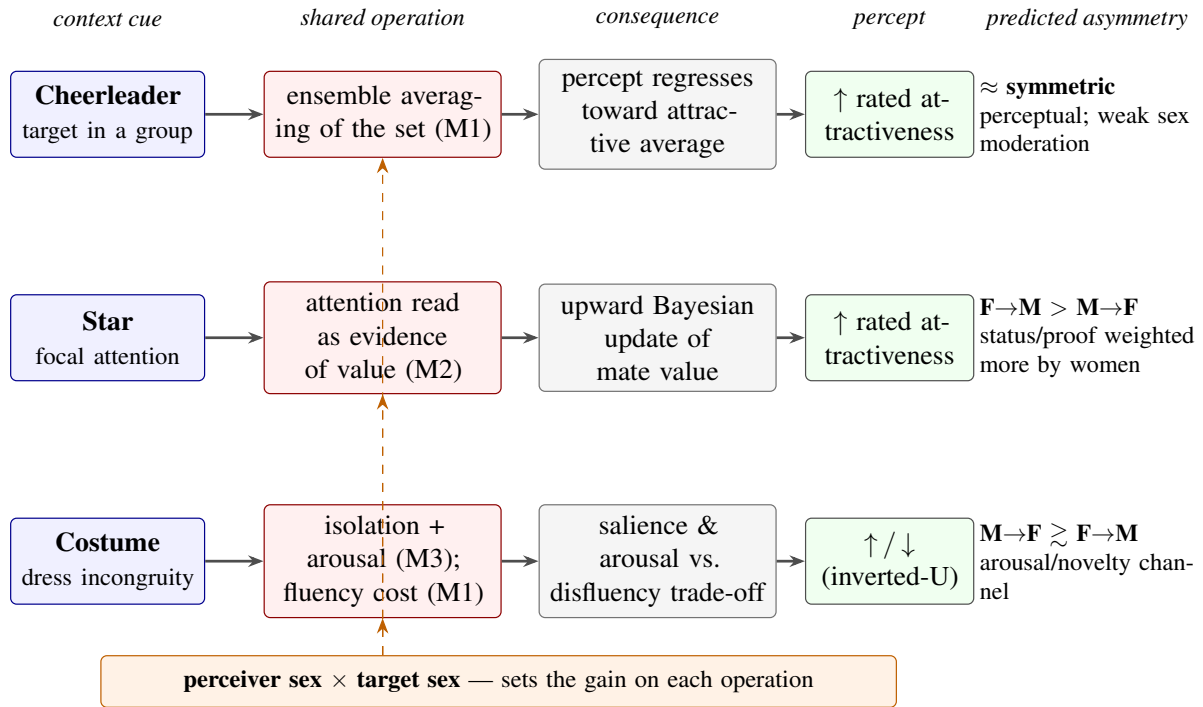


Figure 2. Per-effect mechanism flow (conceptual, not data). Each effect (left) is routed through a shared operation (M1–M3) to a change in rated attractiveness (right); the rightmost column states the predicted cross-sex pattern. The orange bus indicates that perceiver sex × target sex sets the gain on each operation. The costume lane is drawn as inverted-U to mark the predicted reversal under extreme incongruity.

Table 1. Predicted direction and cross-sex pattern for each effect. “Established” refers to the existence of the base phenomenon, not to the present sex-asymmetry predictions, which are novel for all three effects.

Effect	Primary mechanism	Cross-sex prediction	Status
Cheerleader	M1 ensemble perception	$M \rightarrow F \approx F \rightarrow M$ (P2)	Base effect established; mechanism contested
Star	M2 social-proof/status	$F \rightarrow M > M \rightarrow F$ (P3)	Novel framing; components established
Costume	M3 salience × arousal	$M \rightarrow F \gtrsim F \rightarrow M$ (P4)	Novel framing; components established

reverses under extreme incongruity; the cheerleader effect requires simultaneous (or recently remembered) group presentation and may be carried partly by memory (Hsieh et al., 2021).

7 The bottom line: points on a 0–10 scale

The preceding sections answer “*whether* and *why*”. Readers usually also want “*how much*”. Here we give a deliberately concrete—and deliberately illustrative—answer. **Imagine attractiveness as a 0–10 scale on which 5.0 is the (subjective) population average.** Take a target who, seen plainly and alone, would be rated 5.0. How does each effect move that score, and does the move differ for male and female perceivers?

To put effect sizes on this scale we must assume a rating standard deviation. Attractiveness ratings on a 0–10 scale plausibly have $SD \approx 1.5$ points; a standardized effect of Cohen’s d then corresponds to a raw shift of $d \times 1.5$ points. Applying this to the framework’s predicted priors (Figure 3) gives Table 2 and Figure 4.

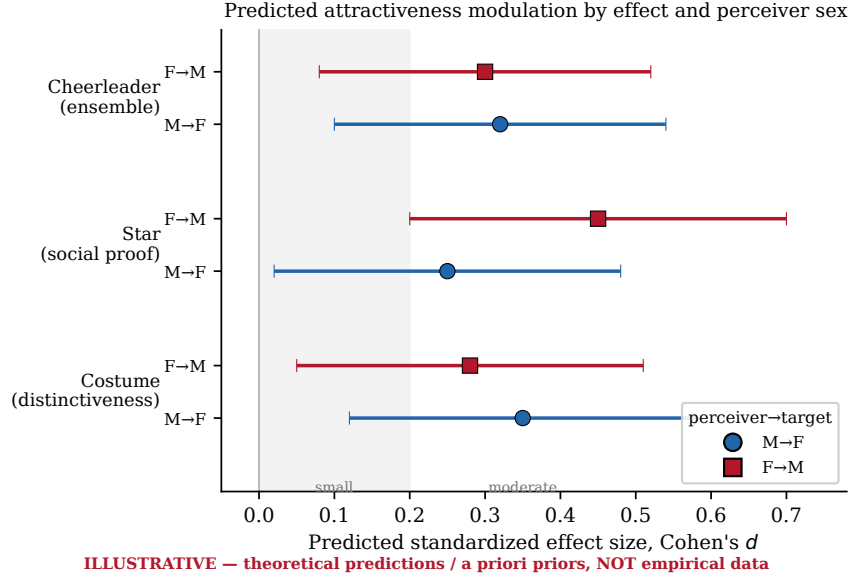


Figure 3. Predicted attractiveness modulation by effect and perceiver sex (illustrative; theoretical predictions, not empirical data). Points are predicted standardized effect sizes (Cohen's d); bars are predicted 95% plausibility intervals. The cheerleader prior is anchored to the small-to-moderate range in the empirical literature (Walker & Vul, 2014); the star and costume priors are conjectures.

Table 2. Illustrative predicted attractiveness of a target who is average (5.0/10) at baseline, after each effect, by perceiver→target direction. Computed as $5.0 + d \times \text{SD}$ with $\text{SD} = 1.5$ and the predicted d of Figure 3. These are model predictions, not measurements; the **+points** column is the predicted lift.

Effect	Male perceiver → female target		Female perceiver → male target	
	predicted d	score (+points)	predicted d	score (+points)
Cheerleader	0.32	5.48 (+0.48)	0.30	5.45 (+0.45)
Star	0.25	5.38 (+0.38)	0.45	5.67 (+0.68)
Costume	0.35	5.53 (+0.53)	0.28	5.42 (+0.42)

The headline answer: **a plainly-average (5/10) person is predicted to be seen as roughly 5.4–5.7/10 once the context effect is added**—a real but modest lift of about half a point, never a transformation. The single biggest mover is the **star effect for women judging men** ($5.0 \rightarrow 5.7$, $+0.7$), consistent with the framework's claim that social-proof/status cues are weighted most heavily in the $F \rightarrow M$ direction. The cheerleader effect is the most symmetric (about $+0.45$ to $+0.48$ either way). The costume effect is larger for men judging women ($+0.53$) than the reverse ($+0.42$), but is the most context-dependent and can reverse under extreme incongruity. We stress three caveats: the $\text{SD} = 1.5$ assumption is a stipulation (a larger SD scales all lifts up proportionally); the underlying d values are conjectured priors, not data; and these are *average* shifts—individual perceivers and targets will vary widely.

8 Illustrative simulations

Caveat (read first). This section reports *simulations*, not experiments. The figures and numbers below are **theoretical predictions generated by toy models**; they are **not** measurements, were **not** obtained from human participants, and should never be cited as empirical estimates.

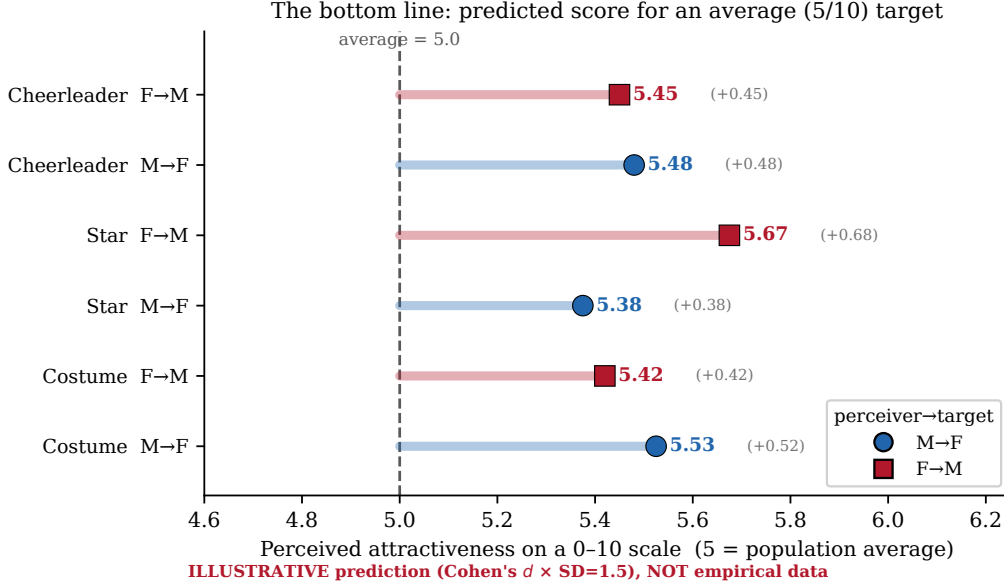


Figure 4. The bottom line (illustrative; not empirical data). Predicted attractiveness, on a 0–10 scale, of a target who is average (5.0) at baseline, under each effect and for each perceiver→target direction. Bars run from the baseline of 5.0 to the predicted score; values are $5.0 + d \times SD$ with $SD = 1.5$. The largest single lift is the star effect for women judging men ($\approx +0.7$ points).

8.1 An ensemble-coding model of the cheerleader effect [ILLUSTRATIVE SIMULATION]

We place faces in a two-dimensional feature space and define attractiveness by *averageness*: a face is more attractive the closer it lies to the population average (Langlois & Roggman, 1990). Ensemble encoding is modelled by biasing each face’s percept toward the mean of its group, $\hat{x}_i = (1 - w)x_i + w\bar{x}_{\text{group}}$, with ensemble weight w . Because a random group’s mean regresses toward the population average (the attractive point), blending members toward that mean tends to raise perceived attractiveness. Across 2×10^4 random four-person groups with $w = 0.45$, the model produces a group-boost for $\approx 90\%$ of faces (Figure 5B). The within-face standardized shift in the model ($d_z \approx 1.25$) is an *internal model quantity* indexing robustness and direction; it is *not* a predicted empirical effect size (the predicted empirical effect is the small-to-moderate d in Figure 3).

8.2 A Bayesian “attention-as-evidence” model of the star effect [ILLUSTRATIVE SIMULATION]

We model an observer inferring a target’s latent mate value V from observed social attention A . With a Gaussian prior $V \sim \mathcal{N}(\mu_0, \sigma_0^2)$ and a likelihood $A \sim \mathcal{N}(V, \sigma_A^2)$ (attention is a noisy read-out of value), the posterior mean is $\mathbb{E}[V | A] = (A/\sigma_A^2 + \mu_0/\sigma_0^2)/(1/\sigma_A^2 + 1/\sigma_0^2)$. Perceived attractiveness tracks this posterior mean. The single sex-dependent parameter is σ_A , the assumed diagnosticity of attention. Encoding the conjecture that women weight social-proof cues more (smaller σ_A for $F \rightarrow M$), the model produces a larger star-boost $F \rightarrow M$ than $M \rightarrow F$ (Figure 6), instantiating P3. The asymmetry is *assumed through* σ_A and then *displayed*; the simulation demonstrates internal consistency, not evidence for the assumption.

9 A proposed empirical programme

We outline three pre-registration-style designs, one per effect, each crossing perceiver sex $\times$ target sex. This is a *forward-looking agenda*: no data have been collected. Stimuli would be drawn from standardized,

ILLUSTRATIVE SIMULATION — theoretical prediction, not empirical data

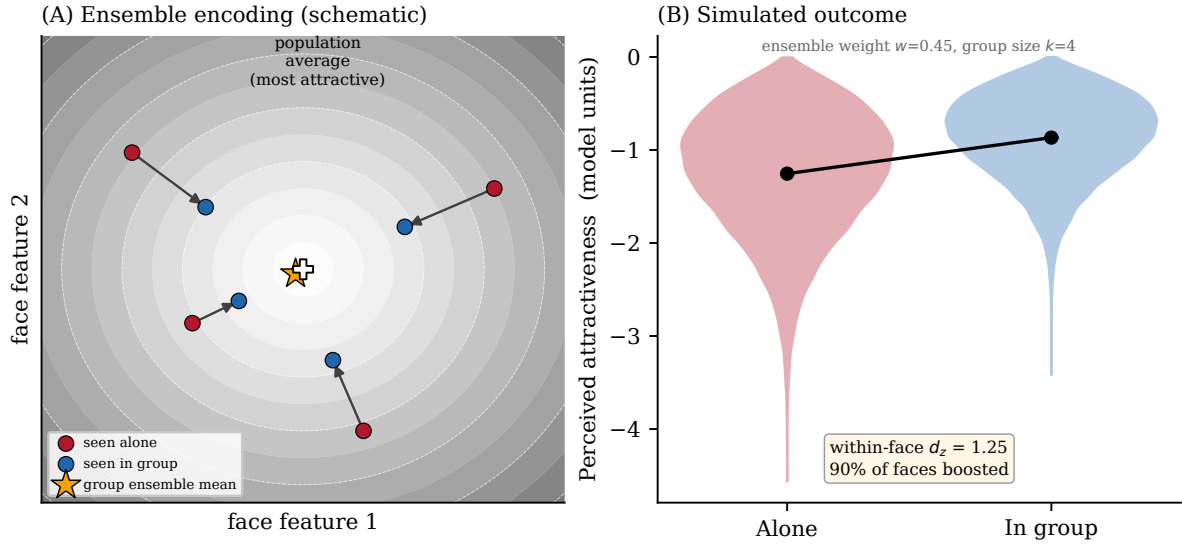


Figure 5. Ensemble-coding account of the cheerleader effect (illustrative simulation; not empirical data). (A) Schematic: in a group, each face’s percept (red→blue arrows) is pulled toward the group ensemble mean (star), which lies nearer the most-attractive population average. (B) Simulated outcome: perceived attractiveness is higher “in group” than “alone”. The reported d_z is an internal model quantity, not a predicted empirical effect size.

consented face/body databases; ratings would use the 0–10 scale of Section 7, with modulated and baseline conditions within perceiver (paired). Power figures assume $\alpha = .05$ two-sided and 80% power for a paired comparison unless noted (Table 3). Because attractiveness rating paradigms are susceptible to demand characteristics and several base effects are replication-fragile (Carragher et al., 2019; Lehmann & Calin-Jageman, 2017), all three designs would (i) use hypothesis-blinded instructions, (ii) include a funnel debrief measuring participants’ guesses as a covariate, and (iii) pre-register the analysis plan and a smallest-effect-size of interest.

Experiment 1 (Cheerleader). *Design.* Each face is rated both alone and embedded in a four-person, same-sex array (order counterbalanced), crossed with target and perceiver sex. *Predictions.* A positive group-boost in both directions with $\Delta_{\text{cheer}}^{\text{M} \rightarrow \text{F}} \approx \Delta_{\text{cheer}}^{\text{F} \rightarrow \text{M}}$ (P2); predicted $d \approx 0.30$ – 0.32 . *Controls.* To adjudicate perception vs. memory (Carragher et al., 2019; Hsieh et al., 2021), include simultaneous vs. sequential presentation; to test the averageness route, manipulate array typicality. *Power.* For $d = 0.30$, $n \approx 90$ perceivers per perceiver-sex cell; the near-zero predicted interaction is treated as an *equivalence* test against a region of practical equivalence (ROPE).

Experiment 2 (Star). *Design.* The same target is shown at the focal centre of social attention (bystanders oriented toward the target; or, in a narrative variant, named as the protagonist) vs. non-focal, crossed with target and perceiver sex. *Predictions.* A positive star boost larger $\text{F} \rightarrow \text{M}$ than $\text{M} \rightarrow \text{F}$ (P3); predicted $d \approx 0.45$ ($\text{F} \rightarrow \text{M}$) and $d \approx 0.25$ ($\text{M} \rightarrow \text{F}$). *Controls.* Hold the target’s pose, dress, and setting constant so only *others’* orientation varies; include an object-attention control to rule out generic gaze-cuing; measure perceived status as a mediator; and explicitly contrast with the Gilovich et al. (2000) spotlight effect by confirming the manipulation shifts *observers’* ratings, not targets’ self-ratings. An optional group-embedding factor would test the predicted cheerleader–star antagonism (P6). *Power.* Powering the smaller $\text{M} \rightarrow \text{F}$ cell ($d = 0.25$) needs $n \approx 128$ per cell; detecting the asymmetry as a between-perceiver interaction ($\Delta d \approx 0.20$) needs ≈ 390 per group—the binding constraint.

ILLUSTRATIVE SIMULATION — theoretical prediction, not empirical data

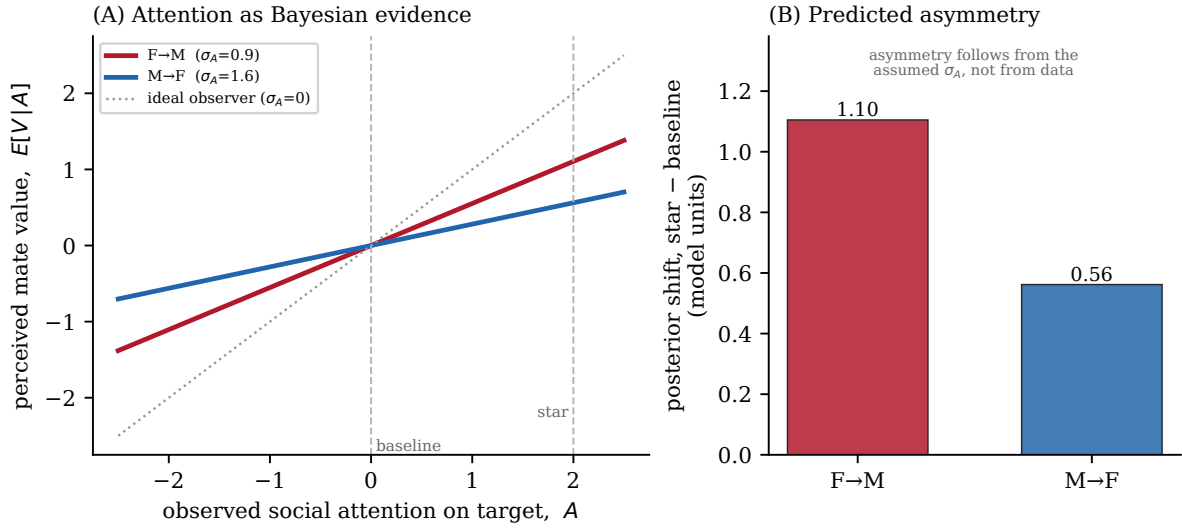


Figure 6. Bayesian attention-as-evidence model of the star effect (illustrative simulation; not empirical data). (A) Posterior mate value rises with observed attention; the slope is steeper when attention is assumed more diagnostic ($F \rightarrow M$). (B) The predicted star “boost” is consequently larger $F \rightarrow M$.

Table 3. Power summary for the proposed experiments ($\alpha = .05$ two-sided, 80% power, paired design). Effect sizes are the framework’s *predicted priors* (illustrative), not observed values.

Effect	Predicted d ($M \rightarrow F$)	Predicted d ($F \rightarrow M$)	n/cell ($M \rightarrow F$)	n/cell ($F \rightarrow M$)	Interaction n/group
Cheerleader	0.32	0.30	79	90	very large (ROPE)
Star	0.25	0.45	128	41	≈ 390
Costume	0.35	0.28	67	103	$\approx 3,200$

Experiment 3 (Costume). *Design.* The same target wears norm-congruent vs. norm-incongruent dress, with incongruity sampled at several levels to test the inverted-U, crossed with target and perceiver sex. *Predictions.* An inverted-U with a peak boost modestly larger $M \rightarrow F$ than $F \rightarrow M$; predicted peak $d \approx 0.35$ ($M \rightarrow F$), 0.28 ($F \rightarrow M$). *Controls.* Hold garment attractiveness and skin exposure constant while varying only norm-congruence; include a colour-red arm (Elliot et al., 2010; Elliot & Niesta, 2008) with the caveat that this effect may not replicate (Lehmann & Calin-Jageman, 2017); collect an arousal measure to test the misattribution route (Dutton & Aron, 1974; Schachter & Singer, 1962). *Power.* For the peak $F \rightarrow M$ cell ($d = 0.28$), $n \approx 103$ per cell; modelling the inverted-U benefits from ≥ 5 incongruity levels.

Powering each *main* effect is feasible (low hundreds), but powering the *cross-sex asymmetries* as interactions is demanding—trivially so for the near-symmetric cheerleader case, and most tractable for the star effect, whose predicted asymmetry is largest. We therefore recommend prioritising the star experiment as the strongest test of the framework’s distinctive prediction.

10 Limitations and ethics

Conceptual limits. The star and costume effects are novel framings; their components are evidenced, but the integrated effects await direct test, and our sex-asymmetry predictions (P3, P4) are conjectures with plausible counter-arguments. Equation 1 is a heuristic scaffold, not a fitted model, and the 0–10 translation (Section 7) inherits its assumptions plus a stipulated rating SD.

The neuroendocrine level is a substrate, not a demonstration. We have linked the three mechanisms to reward circuitry, arousal, and hormones, but these links are imported from adjacent literatures, not shown for these specific context effects. Some of the endocrine evidence is actively contested—most visibly the ovulatory-shift hypothesis, where meta-analyses disagree (Gildersleeve et al., 2014; Wood et al., 2014). The “chemical” level should be read as generating hypotheses (e.g., that the costume effect is arousal-mediated and should be attenuated by an arousal-misattribution control), not as established mechanism.

Sex, sexuality, and the bidirectional frame. We analysed only $M \rightarrow F$ and $F \rightarrow M$ heterosexual judgements, bracketing same-sex attraction, bisexual and asexual perceivers, and non-binary identities, and treating “sex” as a binary design factor. Mate-choice and social-proof mechanisms are not intrinsically heterosexual; a complete account must test whether the predicted asymmetries are about perceiver sex *per se* or about the activated mating goal.

Generalizability. The supporting literature is heavily WEIRD and laboratory-based. Norms of dress, attention, and group display are strongly cultural, so cross-cultural replication is necessary before any claim of universality; the costume effect in particular is *defined* relative to a local norm and cannot be specified culture-free.

Ethics. Research on rated attractiveness raises real concerns: objectification (designs should measure perception without ranking identifiable real people); consent and image rights (consented or synthetic stimuli); representation (findings can be misused to justify appearance-based discrimination, so framing should emphasise that these are context effects on *perception*, not statements of worth); and dual-use (the star and costume manipulations are persuasion techniques). We recommend IRB oversight, pre-registration, and data minimisation.

11 Conclusion

Perceived attractiveness is not a fixed property of a face but a context-sensitive inference. Three context effects—the established cheerleader effect and the novel star and costume effects—are special cases of three mechanisms: ensemble perception, social-proof inference, and salience $\times$ arousal. This version grounded each mechanism biologically (why the cue is informative), neuroendocrinologically (the reward, arousal, and hormonal systems that implement it), and sociologically (status, norms, and proof). The framework predicts approximate sex-symmetry where the mechanism is perceptual (cheerleader), a clear $F \rightarrow M > M \rightarrow F$ asymmetry where it is social-proof (star), and a smaller, non-monotonic $M \rightarrow F \gtrsim F \rightarrow M$ pattern where it is salience $\times$ arousal (costume). Translated to a 0–10 scale, the message is sober: these effects move an average target by roughly half a point, with the star effect lifting a 5.0 man to about 5.7 in women’s eyes—enough to matter at the margin, not enough to remake a face. By locating each effect in a shared mechanistic space we convert a list of curiosities into a small set of parameters whose estimation would tell us not just *that* context moves attractiveness, but *how*, *by how much*, and *for whom*.

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Open science statement

This is a theoretical paper. The two simulations are fully specified by the toy models in Section 8; the figure-generating scripts and a recompile script accompany the manuscript source. No human-subjects data were collected.

References

- Adam, H., & Galinsky, A. D. (2012). Enclothed cognition. *Journal of Experimental Social Psychology*, 48(4), 918–925. doi: 10.1016/j.jesp.2012.02.008
- Aharon, I., Etcoff, N., Ariely, D., Chabris, C. F., O'Connor, E., & Breiter, H. C. (2001). Beautiful faces have variable reward value: fMRI and behavioral evidence. *Neuron*, 32(3), 537–551. doi: 10.1016/S0896-6273(01)00491-3
- Anderson, C., Hildreth, J. A. D., & Howland, L. (2015). Is the desire for status a fundamental human motive? A review of the empirical literature. *Psychological Bulletin*, 141(3), 574–601. doi: 10.1037/a0038781
- Aronson, E., Willerman, B., & Floyd, J. (1966). The effect of a pratfall on increasing interpersonal attractiveness. *Psychonomic Science*, 4(6), 227–228. doi: 10.3758/bf03342263
- Buss, D. M. (1989). Sex differences in human mate preferences: Evolutionary hypotheses tested in 37 cultures. *Behavioral and Brain Sciences*, 12(1), 1–14. doi: 10.1017/S0140525X00023992
- Buss, D. M., & Schmitt, D. P. (1993). Sexual strategies theory: An evolutionary perspective on human mating. *Psychological Review*, 100(2), 204–232. doi: 10.1037/0033-295X.100.2.204
- Carragher, D. J., Thomas, N. A., Gwinn, O. S., & Nicholls, M. E. R. (2019). Limited evidence of hierarchical encoding in the cheerleader effect. *Scientific Reports*, 9(1), 9329. doi: 10.1038/s41598-019-45789-6
- Cialdini, R. B., & Goldstein, N. J. (2004). Social influence: Compliance and conformity. *Annual Review of Psychology*, 55, 591–621. doi: 10.1146/annurev.psych.55.090902.142015
- Cloutier, J., Heatherton, T. F., Whalen, P. J., & Kelley, W. M. (2008). Are attractive people rewarding? Sex differences in the neural substrates of facial attractiveness. *Journal of Cognitive Neuroscience*, 20(6), 941–951. doi: 10.1162/jocn.2008.20062
- Dion, K., Berscheid, E., & Walster, E. (1972). What is beautiful is good. *Journal of Personality and Social Psychology*, 24(3), 285–290. doi: 10.1037/h0033731
- Dutton, D. G., & Aron, A. P. (1974). Some evidence for heightened sexual attraction under conditions of high anxiety. *Journal of Personality and Social Psychology*, 30(4), 510–517. doi: 10.1037/h0037031
- Elliot, A. J., Kayser, D. N., Greitemeyer, T., Lichtenfeld, S., Gramzow, R. H., Maier, M. A., & Liu, H. (2010). Red, rank, and romance in women viewing men. *Journal of Experimental Psychology: General*, 139(3), 399–417. doi: 10.1037/a0019689
- Elliot, A. J., & Niesta, D. (2008). Romantic red: Red enhances men's attraction to women. *Journal of Personality and Social Psychology*, 95(5), 1150–1164. doi: 10.1037/0022-3514.95.5.1150
- Gildersleeve, K., Haselton, M. G., & Fales, M. R. (2014). Do women's mate preferences change across the ovulatory cycle? A meta-analytic review. *Psychological Bulletin*, 140(5), 1205–1259. doi: 10.1037/a0035438
- Gilovich, T., Medvec, V. H., & Savitsky, K. (2000). The spotlight effect in social judgment: An egocentric bias in estimates of the salience of one's own actions and appearance. *Journal of Personality and Social Psychology*, 78(2), 211–222. doi: 10.1037/0022-3514.78.2.211
- Gouda-Vossos, A., Nakagawa, S., Dixon, B. J. W., & Brooks, R. C. (2018). Mate choice copying in humans: A systematic review and meta-analysis. *Adaptive Human Behavior and Physiology*, 4(4), 364–386. doi: 10.1007/s40750-018-0099-y
- Haberman, J., & Whitney, D. (2007). Rapid extraction of mean emotion and gender from sets of faces. *Current Biology*, 17(17), R751–R753. doi: 10.1016/j.cub.2007.06.039

- Henrich, J., & Gil-White, F. J. (2001). The evolution of prestige: Freely conferred deference as a mechanism for enhancing the benefits of cultural transmission. *Evolution and Human Behavior*, 22(3), 165–196. doi: 10.1016/S1090-5138(00)00071-4
- Hsieh, J. Y. J., Gwinn, O. S., Brooks, K. R., Stephen, I. D., Carragher, D. J., & Nicholls, M. E. R. (2021). The “cheerleader effect” in facial and bodily attractiveness: A result of memory bias and not perceptual encoding. *Quarterly Journal of Experimental Psychology*, 74(5), 972–980. doi: 10.1177/1747021820976087
- Hunt, R. R. (1995). The subtlety of distinctiveness: What von Restorff really did. *Psychonomic Bulletin & Review*, 2(1), 105–112. doi: 10.3758/bf03214414
- Kampe, K. K. W., Frith, C. D., Dolan, R. J., & Frith, U. (2001). Reward value of attractiveness and gaze. *Nature*, 413(6856), 589. doi: 10.1038/35098149
- Langlois, J. H., Kalakanis, L., Rubenstein, A. J., Larson, A., Hallam, M., & Smoot, M. (2000). Maxims or myths of beauty? a meta-analytic and theoretical review. *Psychological Bulletin*, 126(3), 390–423. doi: 10.1037/0033-2909.126.3.390
- Langlois, J. H., & Roggman, L. A. (1990). Attractive faces are only average. *Psychological Science*, 1(2), 115–121. doi: 10.1111/j.1467-9280.1990.tb00079.x
- Lehmann, G. K., & Calin-Jageman, R. J. (2017). Is red really romantic? two pre-registered replications of the red-romance hypothesis. *Social Psychology*, 48(3), 174–183. doi: 10.1027/1864-9335/a000296
- Little, A. C., Burriss, R. P., Jones, B. C., DeBruine, L. M., & Caldwell, C. A. (2008). Social influence in human face preference: Men and women are influenced more for long-term than short-term attractiveness decisions. *Evolution and Human Behavior*, 29(2), 140–146. doi: 10.1016/j.evolhumbehav.2007.11.007
- Little, A. C., Jones, B. C., & DeBruine, L. M. (2011). Facial attractiveness: Evolutionary based research. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 366(1571), 1638–1659. doi: 10.1098/rstb.2010.0404
- Mazur, A., & Booth, A. (1998). Testosterone and dominance in men. *Behavioral and Brain Sciences*, 21(3), 353–363. doi: 10.1017/S0140525X98001228
- O’Doherty, J., Winston, J., Critchley, H., Perrett, D., Burt, D. M., & Dolan, R. J. (2003). Beauty in a smile: The role of medial orbitofrontal cortex in facial attractiveness. *Neuropsychologia*, 41(2), 147–155. doi: 10.1016/S0028-3932(02)00145-8
- Place, S. S., Todd, P. M., Penke, L., & Asendorpf, J. B. (2010). Humans show mate copying after observing real mate choices. *Evolution and Human Behavior*, 31(5), 320–325. doi: 10.1016/j.evolhumbehav.2010.02.001
- Rhodes, G. (2006). The evolutionary psychology of facial beauty. *Annual Review of Psychology*, 57, 199–226. doi: 10.1146/annurev.psych.57.102904.190208
- Roney, J. R., Hanson, K. N., Durante, K. M., & Maestripieri, D. (2006). Reading men’s faces: Women’s mate attractiveness judgments track men’s testosterone and interest in infants. *Proceedings of the Royal Society B: Biological Sciences*, 273(1598), 2169–2175. doi: 10.1098/rspb.2006.3569
- Schachter, S., & Singer, J. (1962). Cognitive, social, and physiological determinants of emotional state. *Psychological Review*, 69(5), 379–399. doi: 10.1037/h0046234
- Scheele, D., Wille, A., Kendrick, K. M., Stoffel-Wagner, B., Becker, B., Güntürkün, O., . . . Hurlemann, R. (2013). Oxytocin enhances brain reward system responses in men viewing the face of their female partner. *Proceedings of the National Academy of Sciences*, 110(50), 20308–20313. doi: 10.1073/pnas.1314190110
- Thorndike, E. L. (1920). A constant error in psychological ratings. *Journal of Applied Psychology*, 4(1), 25–29. doi: 10.1037/h0071663
- Thornhill, R., & Gangestad, S. W. (1999). Facial attractiveness. *Trends in Cognitive Sciences*, 3(12), 452–460. doi: 10.1016/S1364-6613(99)01403-5
- Trivers, R. L. (1972). Parental investment and sexual selection. In B. Campbell (Ed.), *Sexual selection and the descent of man, 1871–1971* (pp. 136–179). Chicago: Aldine.
- van Osch, Y., Blanken, I., Meijs, M. H. J., & van Wolferen, J. (2015). A group’s physical attractiveness is

- greater than the average attractiveness of its members: The group attractiveness effect. *Personality and Social Psychology Bulletin*, 41(4), 559–574. doi: 10.1177/0146167215572799
- Walker, D., & Vul, E. (2014). Hierarchical encoding makes individuals in a group seem more attractive. *Psychological Science*, 25(1), 230–235. doi: 10.1177/0956797613497969
- Waynforth, D. (2007). Mate choice copying in humans. *Human Nature*, 18(3), 264–271. doi: 10.1007/s12110-007-9004-2
- Webster, M., Jr., & Driskell, J. E., Jr. (1983). Beauty as status. *American Journal of Sociology*, 89(1), 140–165. doi: 10.1086/227836
- Whitney, D., & Yamanashi Leib, A. (2018). Ensemble perception. *Annual Review of Psychology*, 69, 105–129. doi: 10.1146/annurev-psych-010416-044232
- Winston, J. S., O’Doherty, J., Kilner, J. M., Perrett, D. I., & Dolan, R. J. (2007). Brain systems for assessing facial attractiveness. *Neuropsychologia*, 45(1), 195–206. doi: 10.1016/j.neuropsychologia.2006.05.009
- Wood, W., Kressel, L., Joshi, P. D., & Louie, B. (2014). Meta-analysis of menstrual cycle effects on women’s mate preferences. *Emotion Review*, 6(3), 229–249. doi: 10.1177/1754073914523073
- Ying, H., Burns, E. J., Lin, X., & Xu, H. (2019). Ensemble statistics shape face adaptation and the cheerleader effect. *Journal of Experimental Psychology: General*, 148(3), 421–436. doi: 10.1037/xge0000564